

Tutorial 6: Converters
EE 3203: Power Electronics

1. Fig. 1 shows a Bridge Converter. Assuming continuous conduction,
 - a. Apply the principle of volt-sec balance to find the relationship between input (V_i) and output (V_o) average voltages.
 - b. Sketch the voltage across the inductor for a steady state duty ratio, $D = 0.25$.
 - c. Plot the average output voltage (V_o) curve for $0 \leq D \leq 1$.

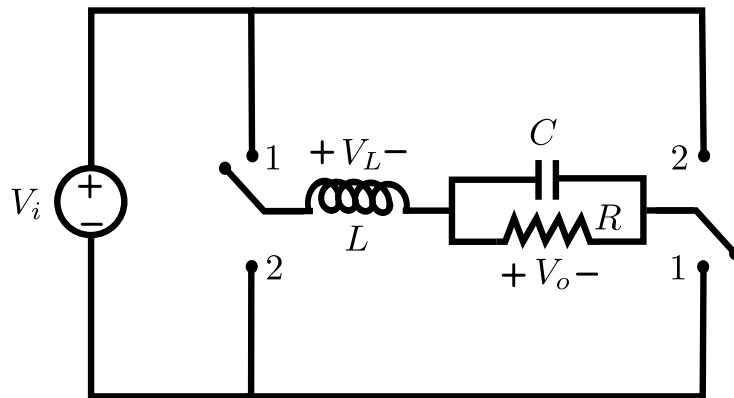


Fig. 1: Bridge Converter

2. Fig. 2 shows a Watkins-Johnson Converter. Assuming continuous conduction,

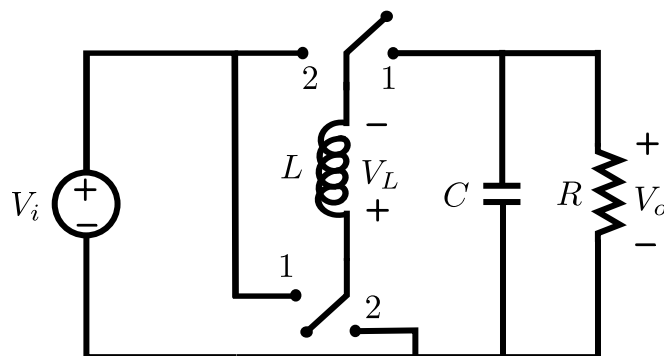


Fig. 2: Watkins-Johnson Converter

- a. Apply the principle of volt-sec balance to find the relationship between input (V_i) and output (V_o) average voltages.
- b. Sketch the voltage across the inductor for a steady state duty ratio, $D = 0.25$.

- c. Plot the average output voltage (V_o) curve for $0 \leq D \leq 1$.
3. Fig. 3 shows a Buck ² Converter. Assuming continuous conduction,
- Apply the principle of volt-sec balance to find the relationship between input (V_i) and output (V_o) average voltages.
 - Sketch the voltage across the two inductors for a steady state duty ratio, $D = 0.25$.
 - Plot the average output voltage (V_o) curve for $0 \leq D \leq 1$.

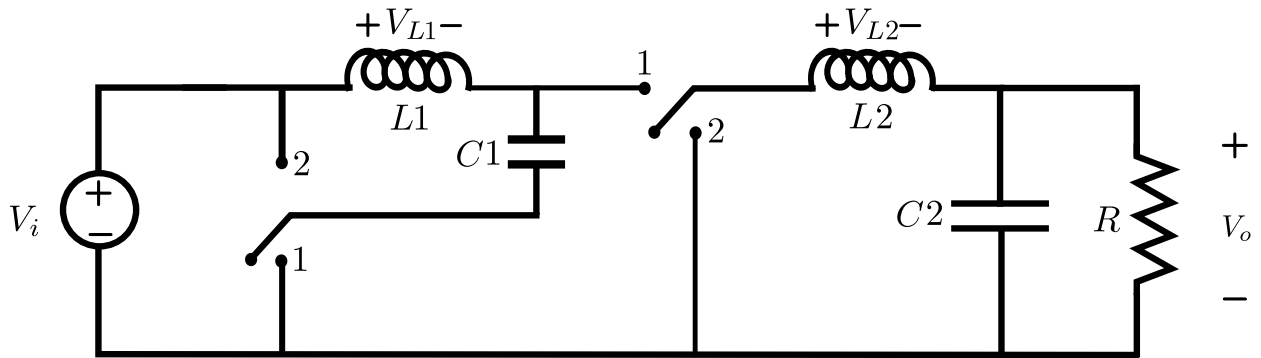


Fig. 3: Buck ² Converter